

Continuous Biomanufacturing of Mabs: Challenges Future Trends and Significance of Cost Modeling

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Agenda

Enzene's Integrated Continuous Bioprocess

Introduction: Process Intensification

Value for Process and COG's modeling

Continuous Process vs Fed Batch: Cost Analysis

Understanding Challenges and Cost Implications

Summary



Lets talk Costs \$\$

Cost of Gold per Gram

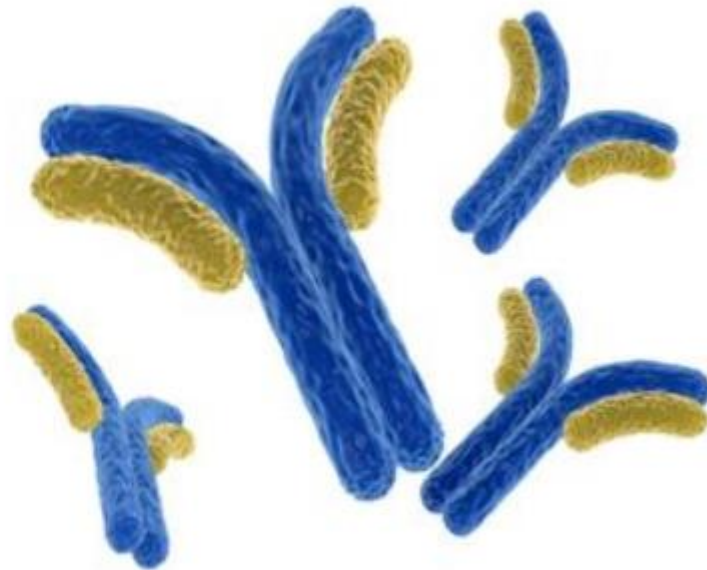
\$ 50



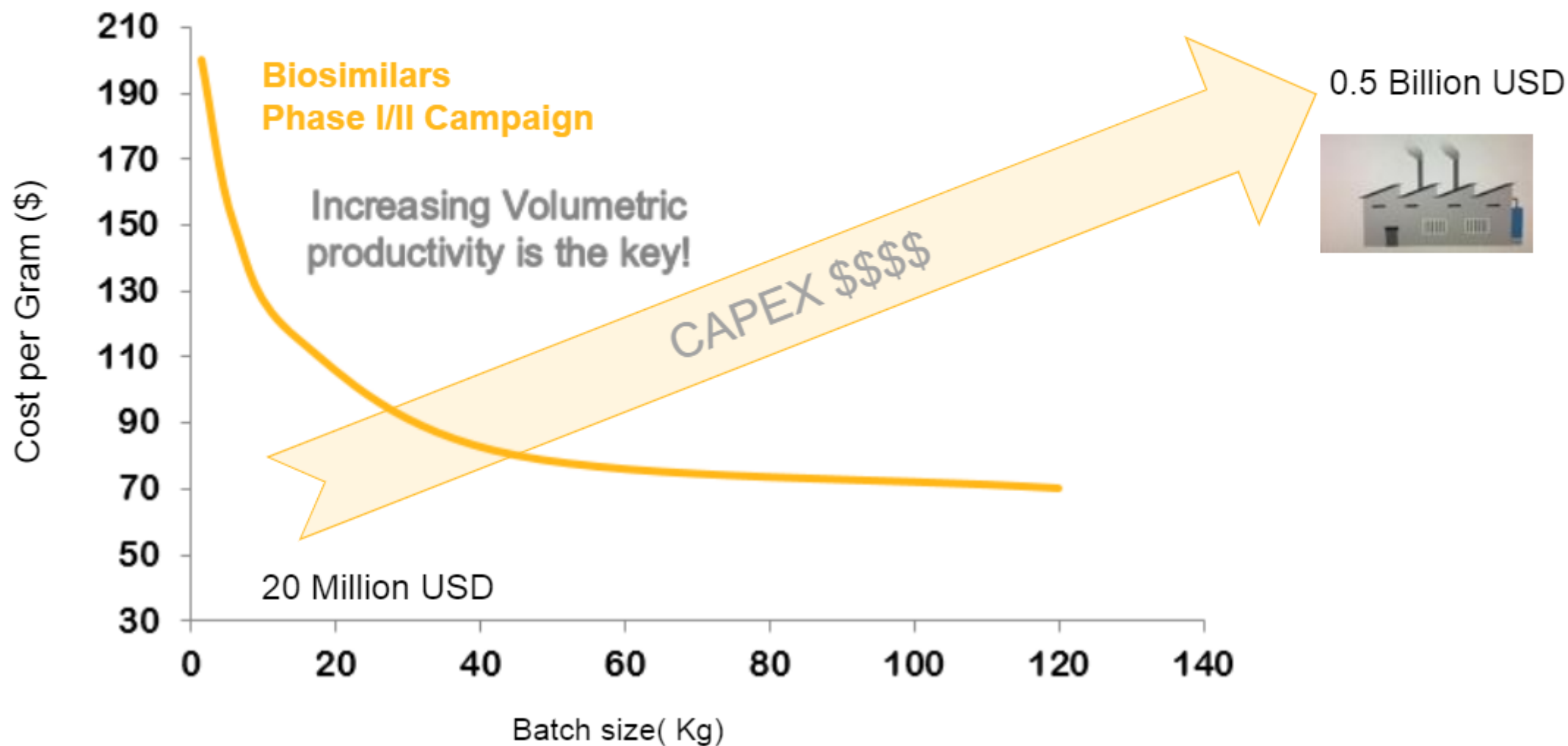
Lets talk Costs \$\$

Cost of mAb production per Gram

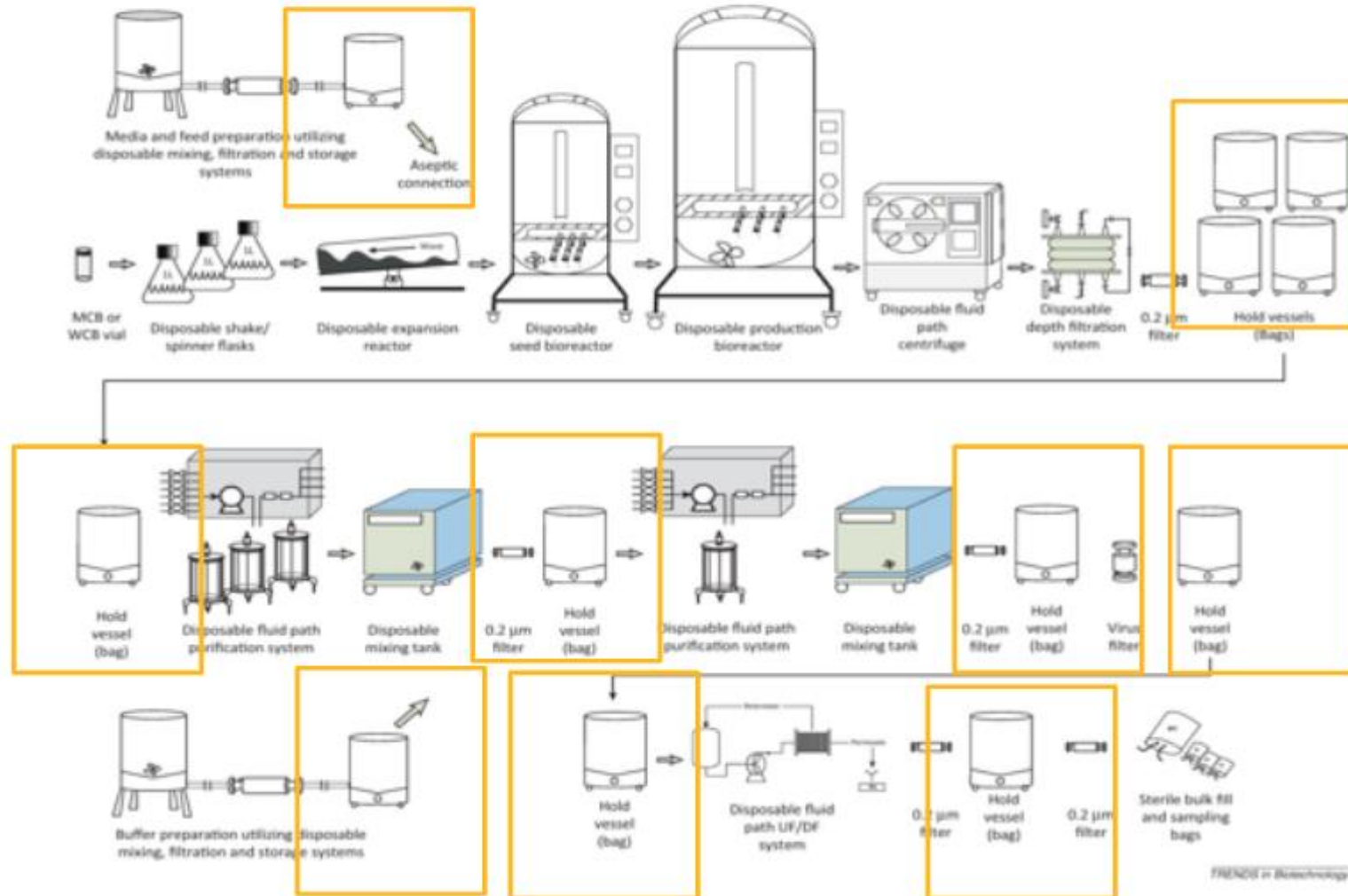
\$ 150



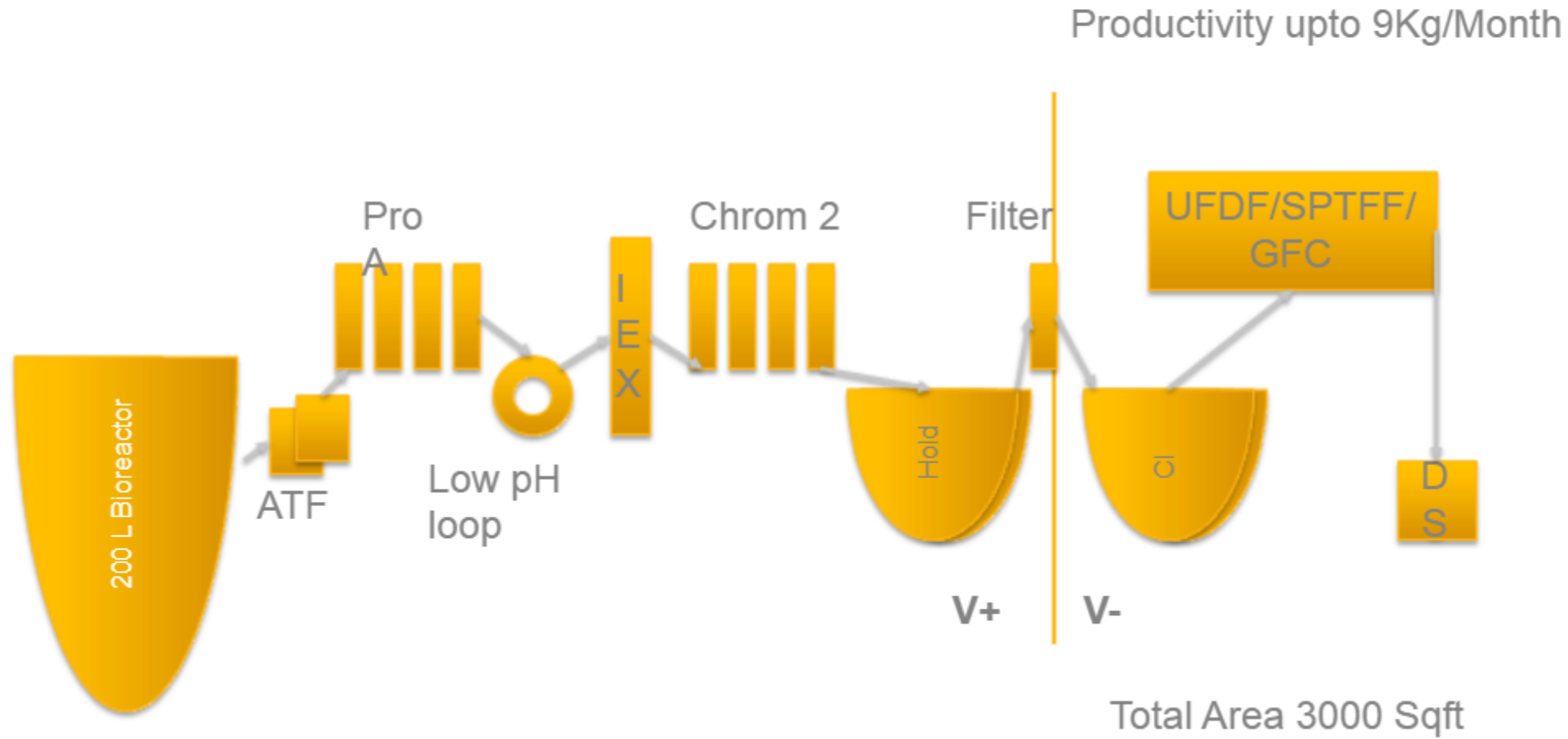
The magic number \$ 50/g



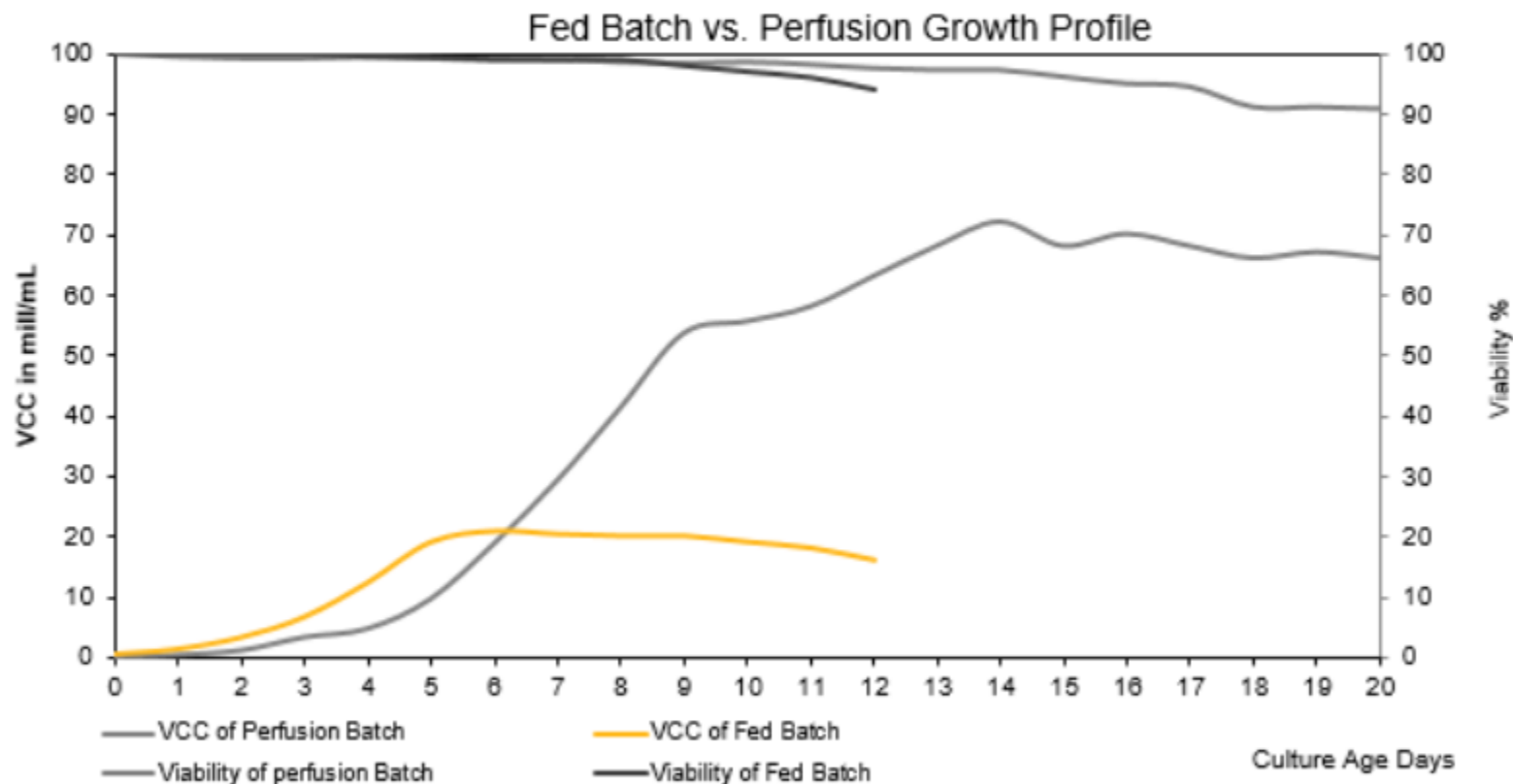
Fed-Batch Plants are Clunky!



Continuous Process

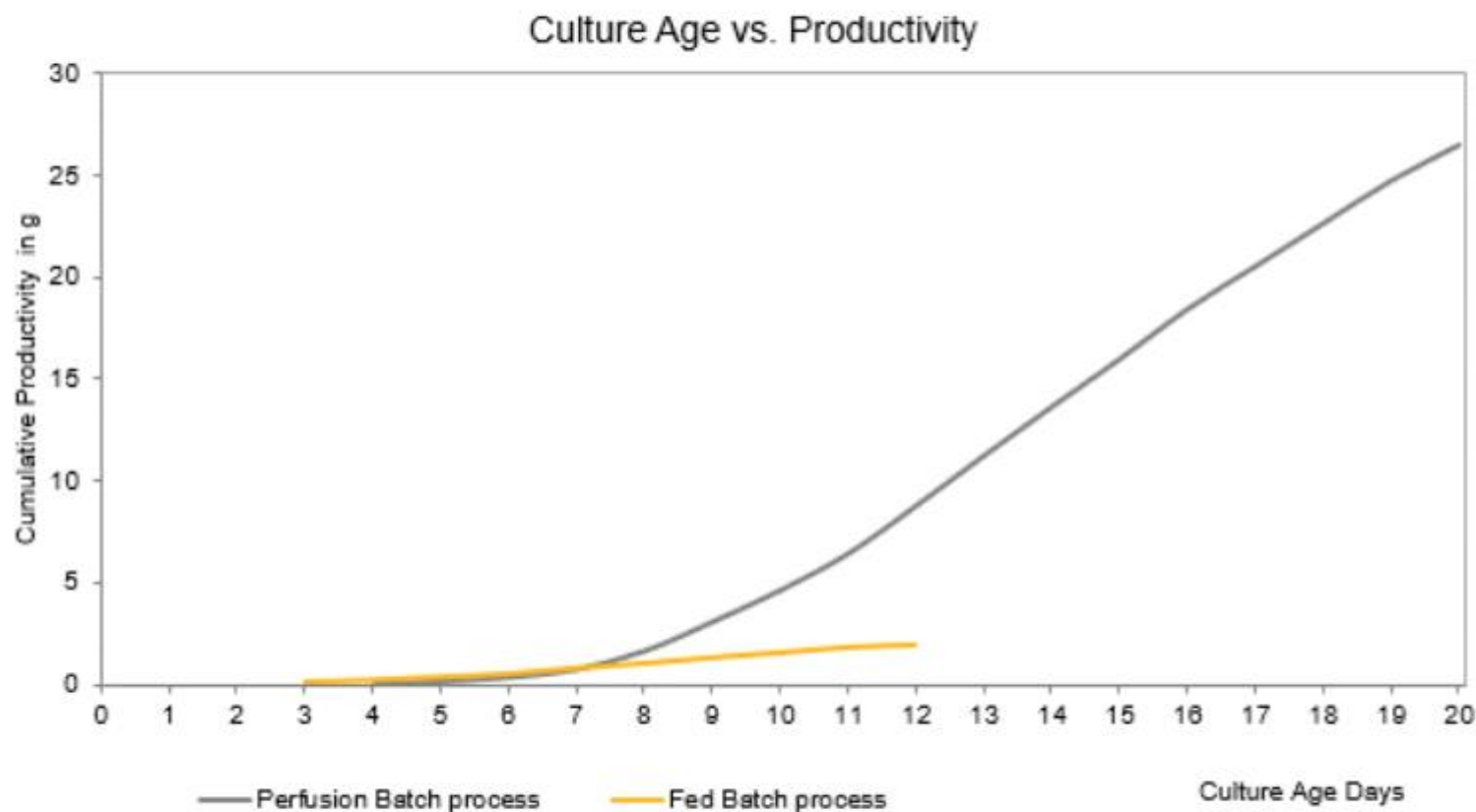


Growth Profile Fed Batch vs Perfusion



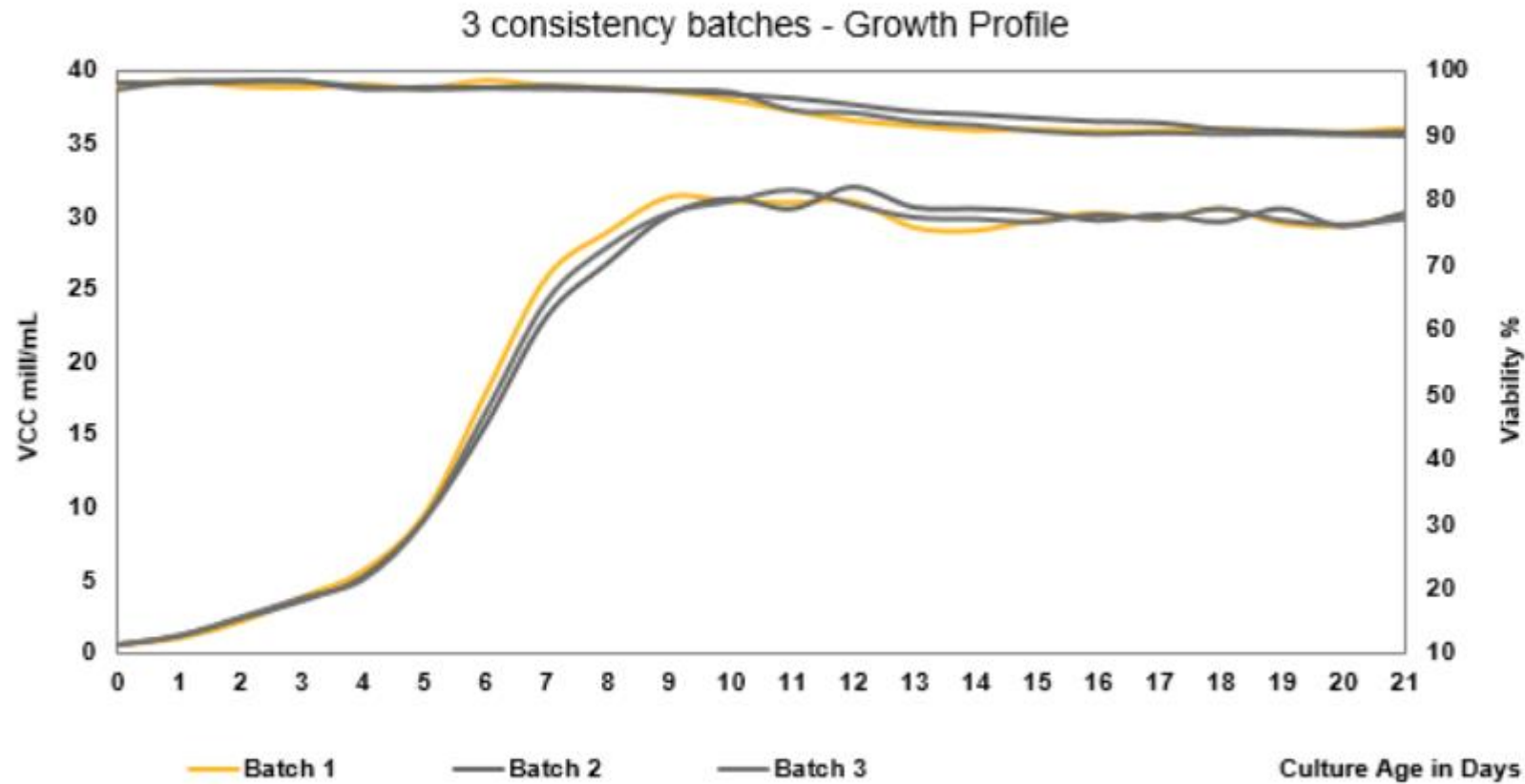
Comparison of Fed Batch (12 Days) vs. Perfusion (20 Days) at 2L scale

Productivity Fed Batch vs Perfusion

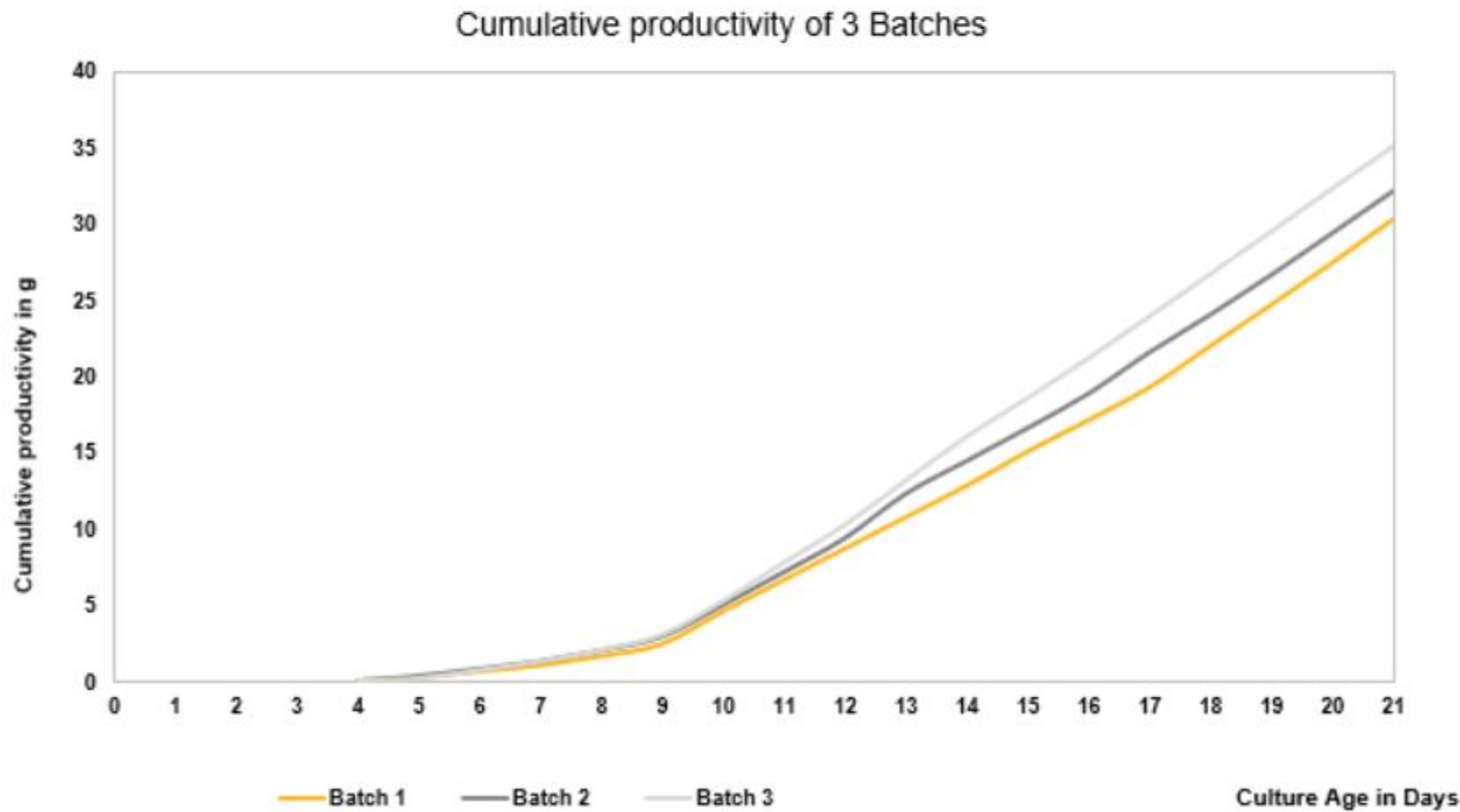


Productivity @ 2L scale in Fed batch is 3.2 g (12 days) and In Perfusion 27 g (20days)

Process Consistency

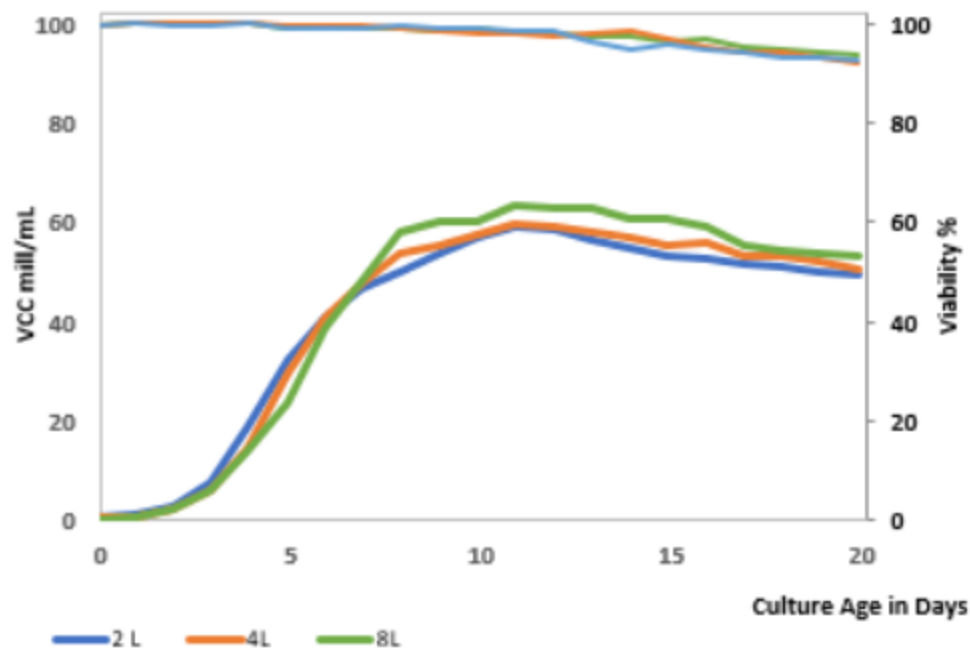


Process Consistency

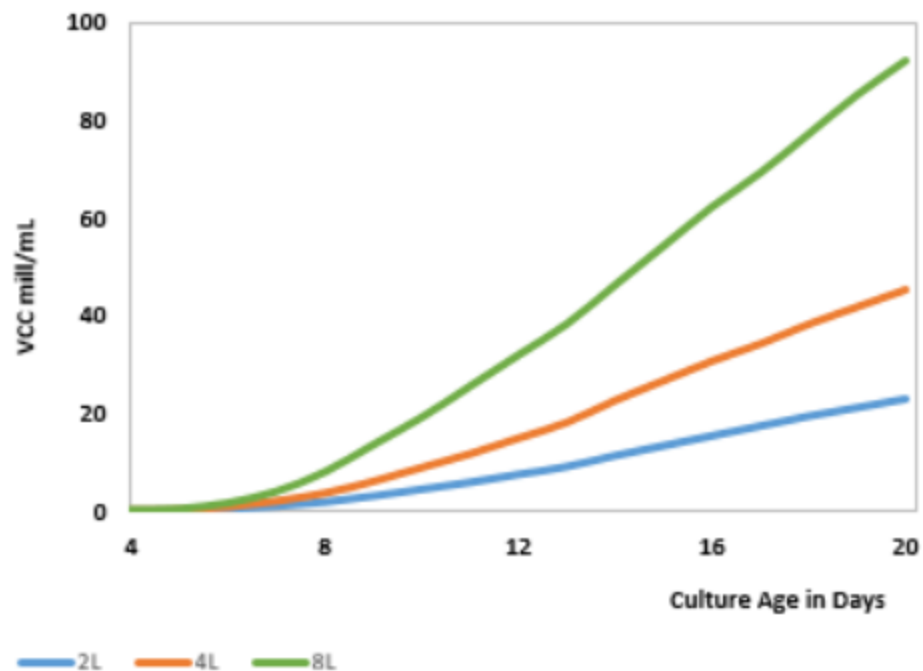


Process scalability

Growth Profile (2L,4L & 8L)

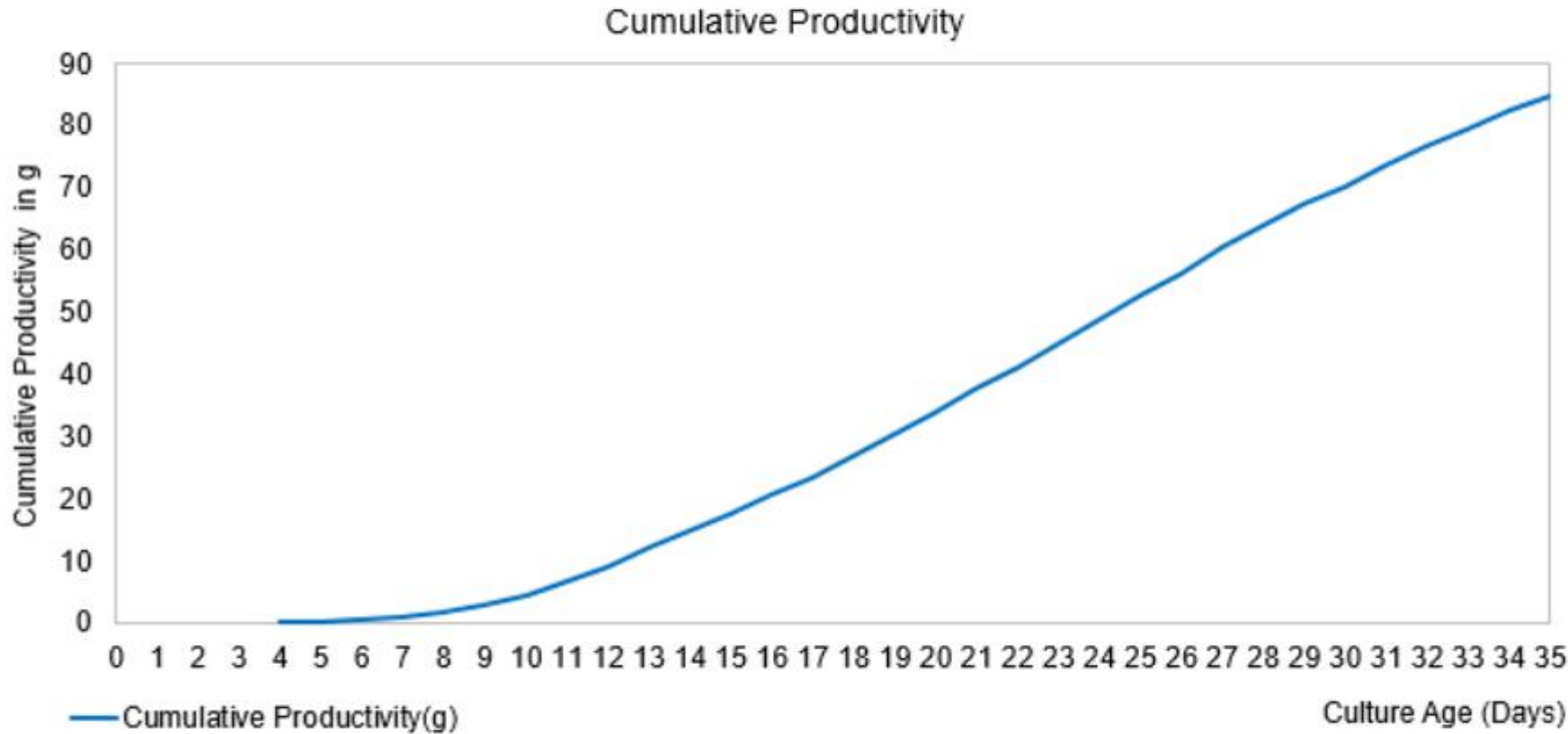


Cumulative Productivity(g) (2L,4L & 8L)



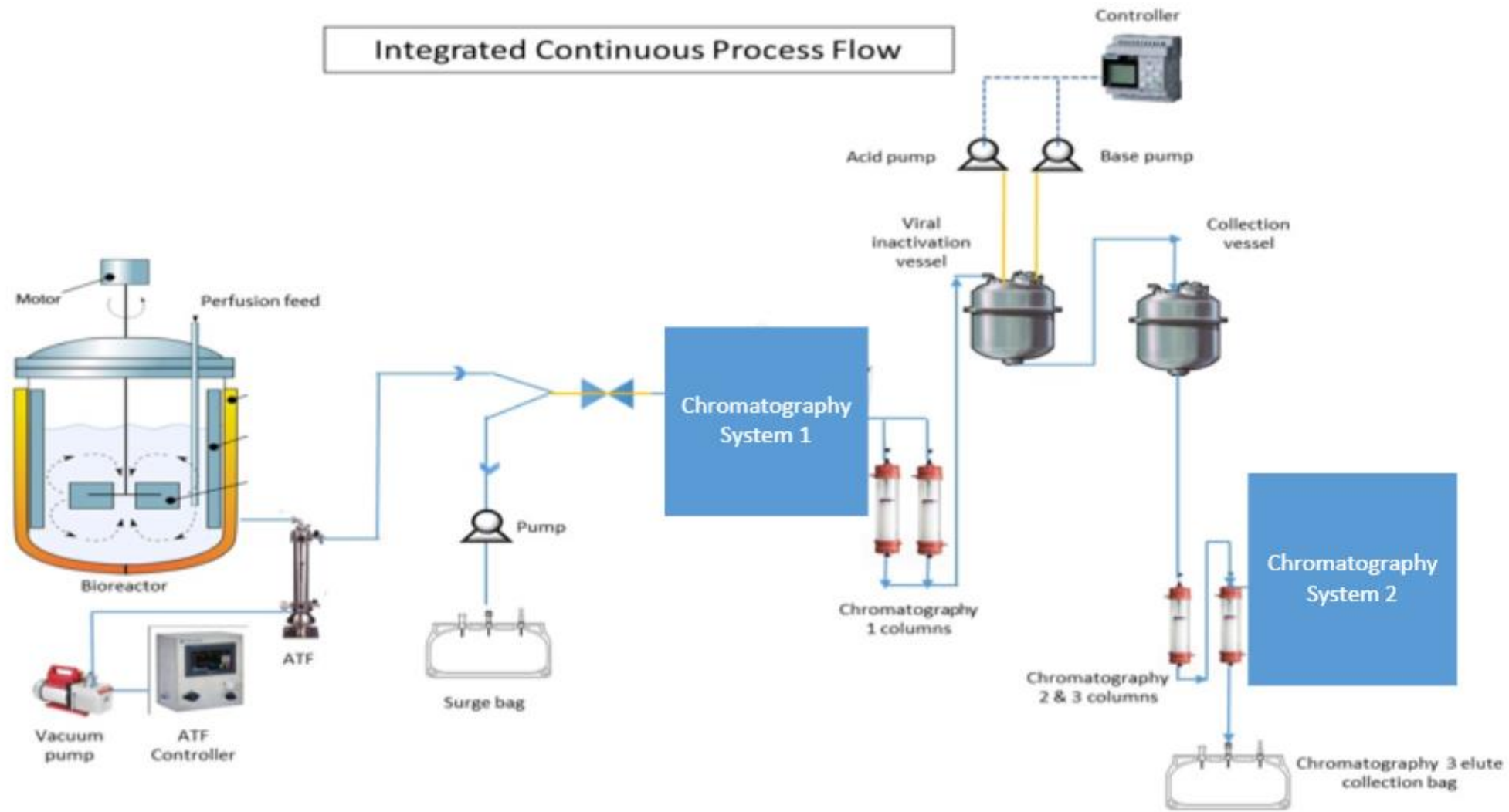
2L scale: 22.5 g
4L scale-: 45 g
8L scale: 92.4 g

Cumulative Productivity



- Productivity in a 2L scale is 43g/L over 35 days

Integrated Continuous Process Flow

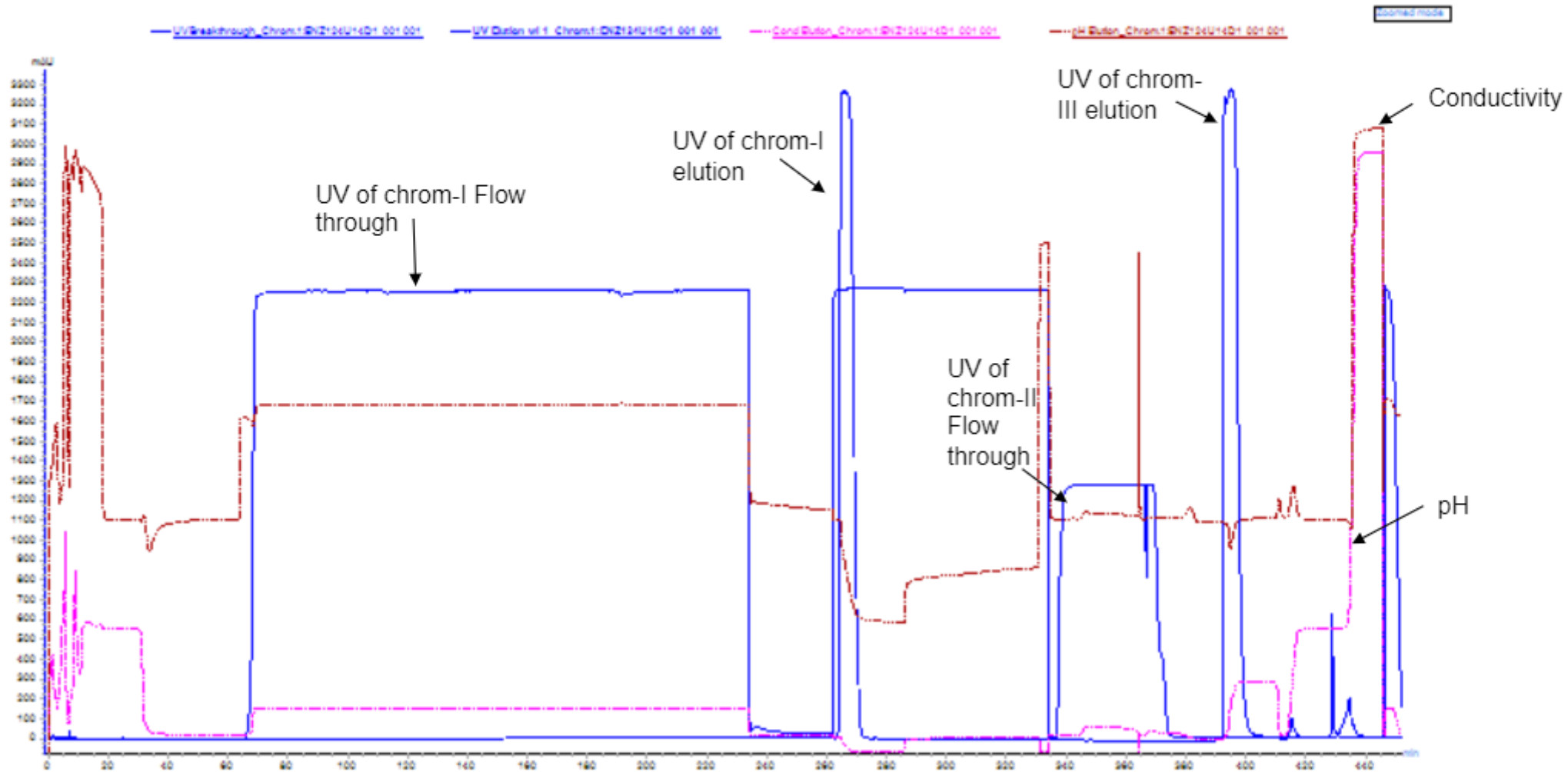


Pilot GMP (Enzene's Unit II)

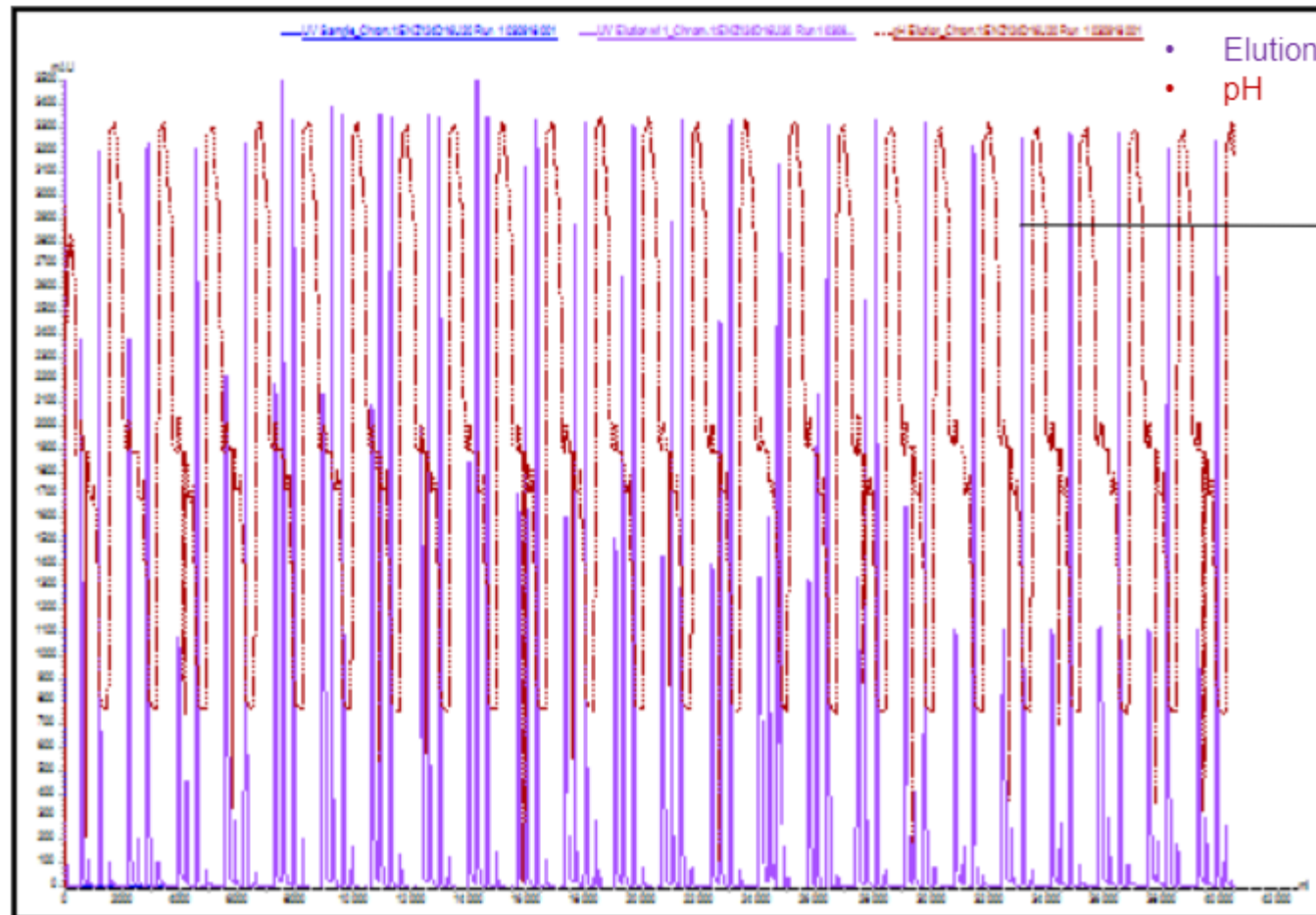


Proof of concept for fully continuous is established at 50 L bioreactor scale

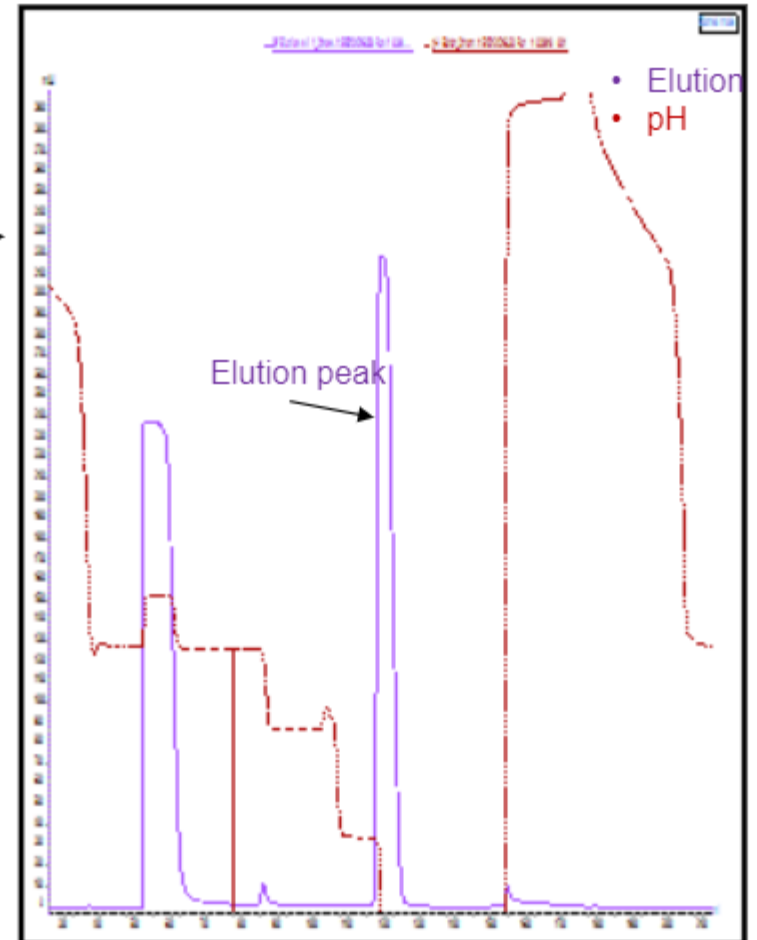
Representative chromatogram



Chromatography 1

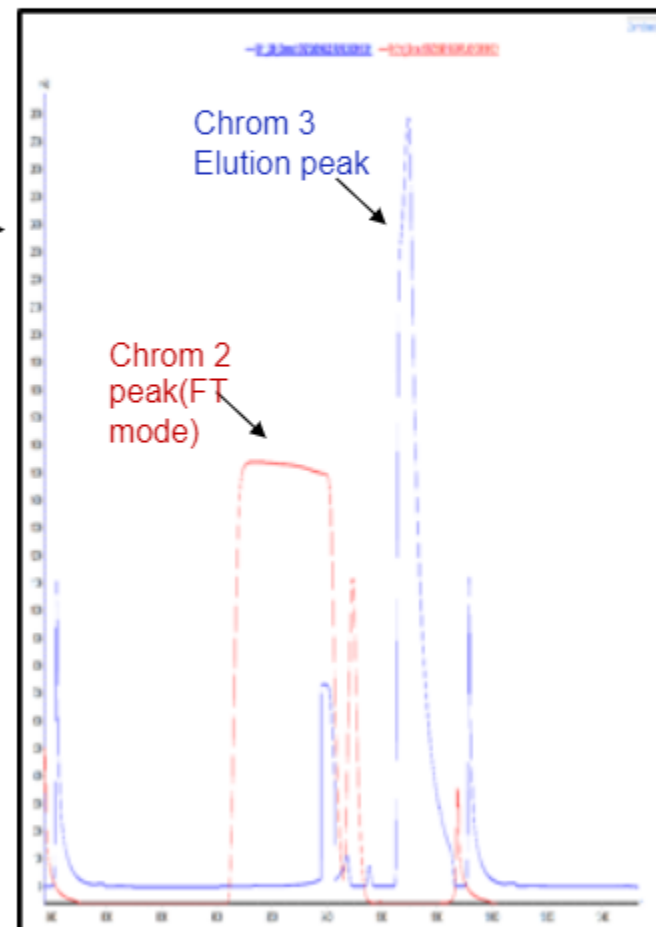
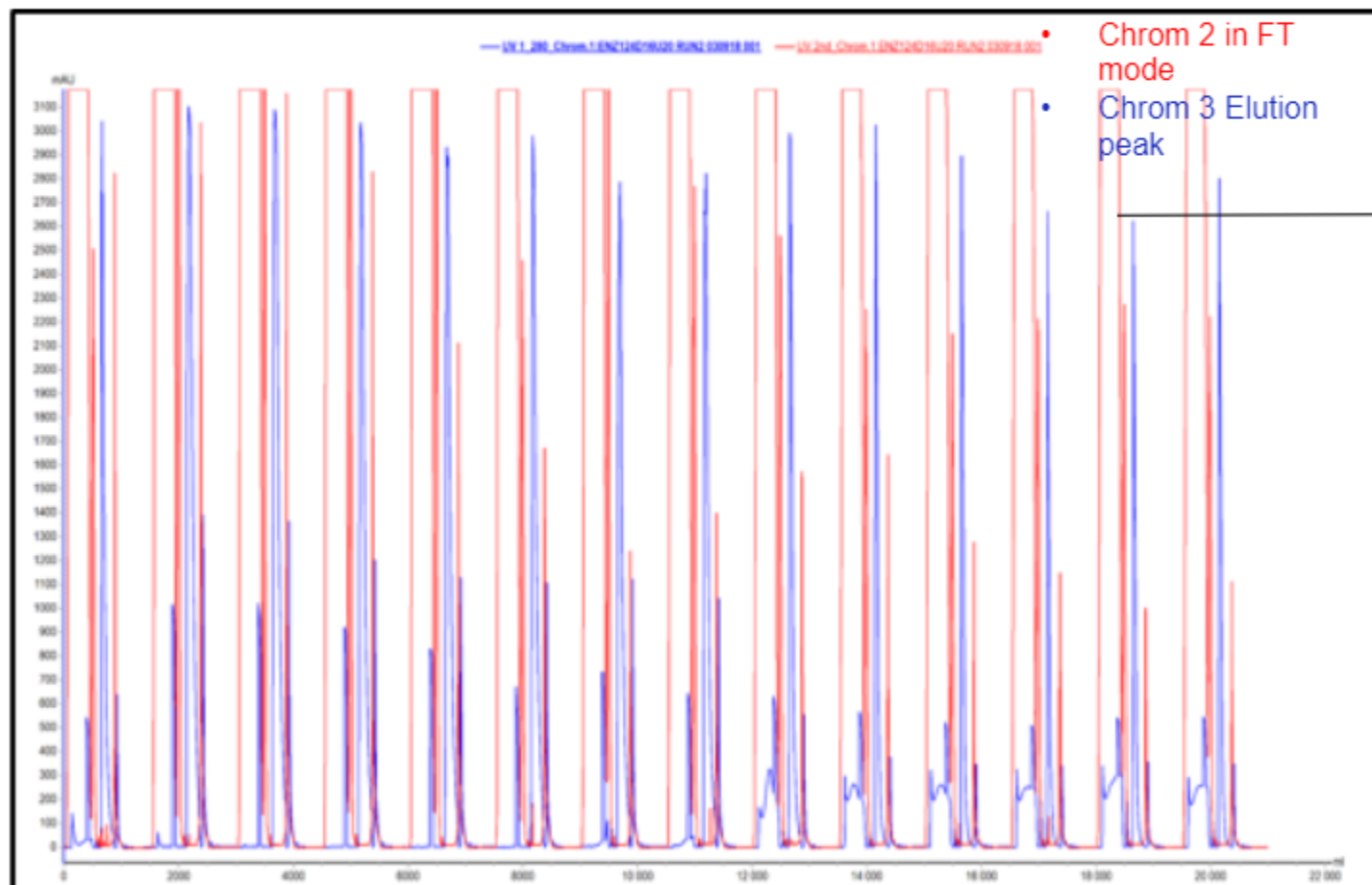


Chromatography 1 : Performed in continuous mode



Zoomed profile of chromatography 1

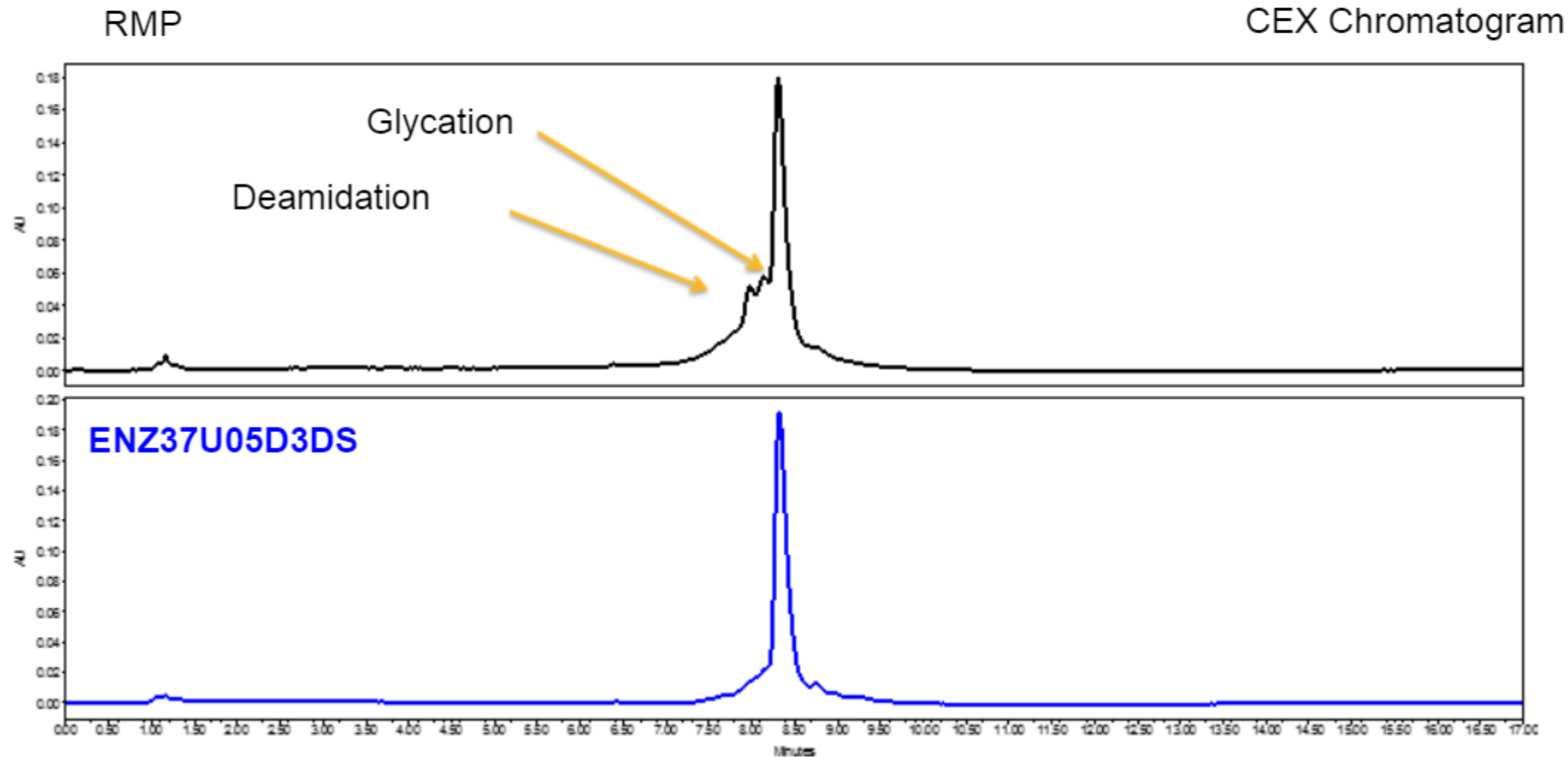
Chromatography 2 & 3



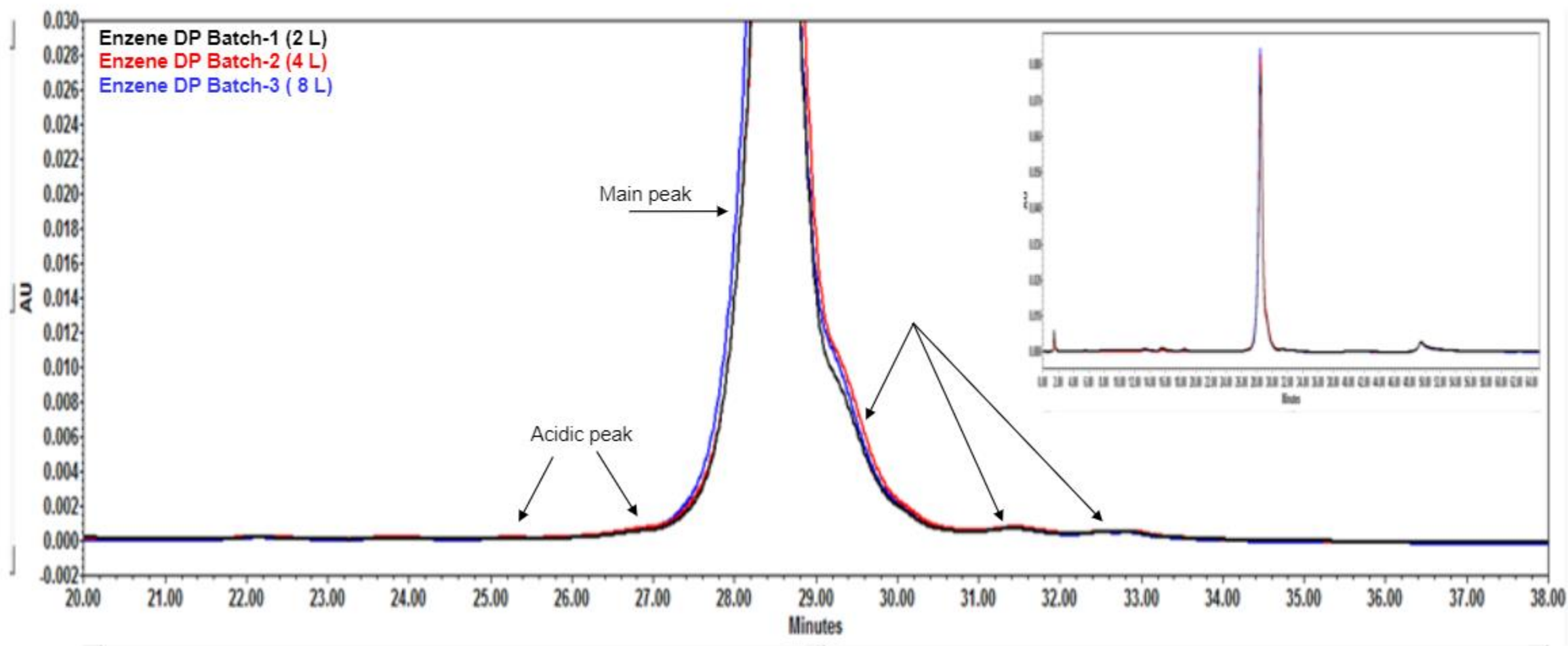
Chromatography 2&3 : Performed in continuous mode

Zoomed profile of chromatography 2&3

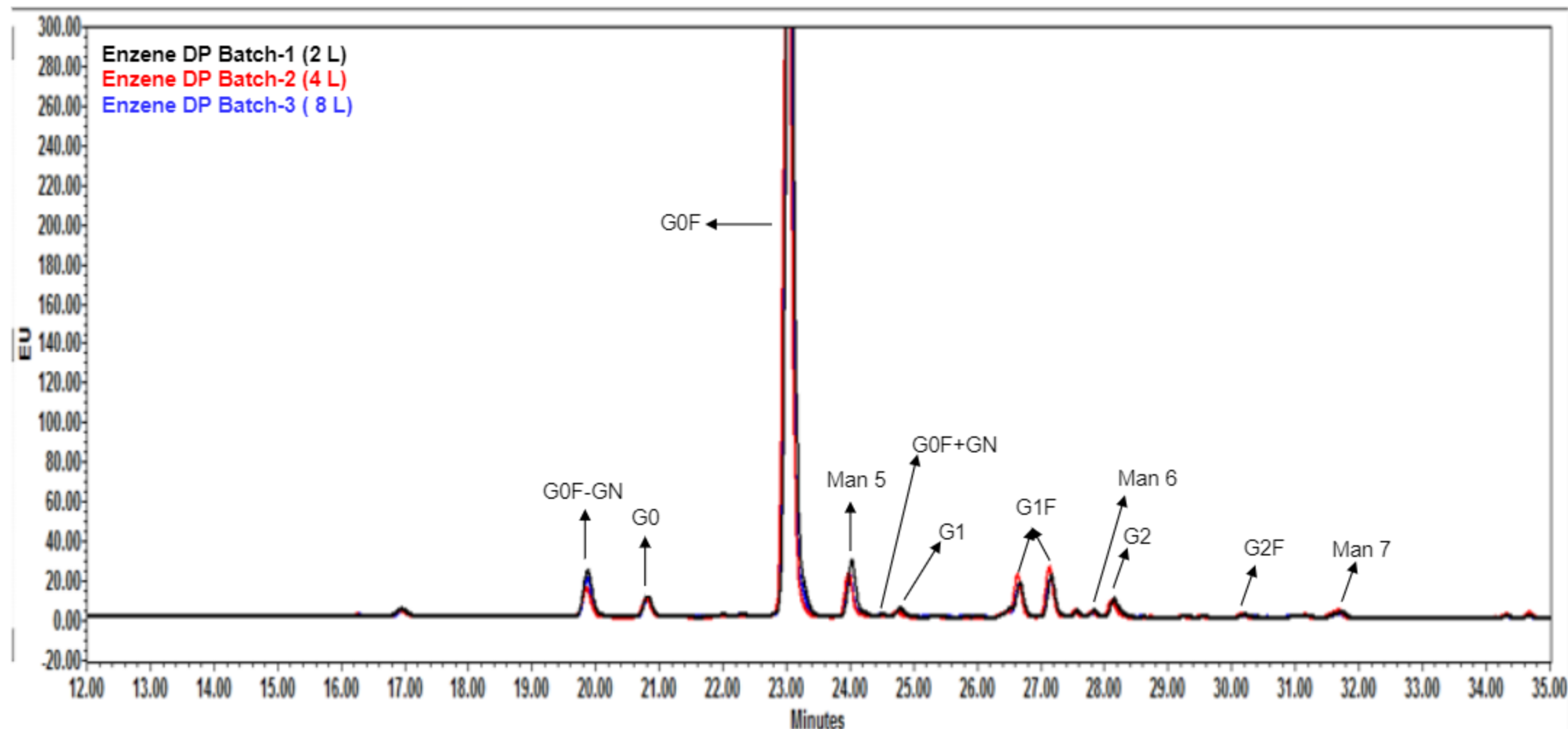
Consistency in product quality across the three different batches



Consistency in product quality across the three different batches



Consistency in product quality across the three different batches



Unit-1



Enzene Opens its First Continuous Biologics Manufacturing Facility With a Promise to Disrupt the mAb Manufacturing Cost



NEWS PROVIDED BY
Enzene Biosciences Ltd. →
07 Nov, 2019, 17:54 IST

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PUNE, India, Nov. 7, 2019 /PRNewswire/ -- Enzene Biosciences today formally opened its first fully connected continuous Biologics manufacturing facility in Pune, India. This first-of-its-kind facility was built in half the time as compared to the conventional biologics manufacturing plants. The state-of-the-art manufacturing plant has a significantly smaller footprint than the conventional manufacturing facilities.

Enzene is amongst the first movers in end-to-end connected bio manufacturing to have set up a fully automated continuous cGMP compliant manufacturing plant for mAb production.



Enzenes Unit 1 Chakan

This plant as compared to a traditional manufacturing plant is capable of delivering higher production, and will also enable Enzene as well its potential clients to rapidly move their early/pre-clinical assets to the later stage of development or to the commercial stage.

Agenda

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Manufacturing Productivity Beyond Initial Facility Capabilities

Value of extra batches per year? – SU vs SS

Value of productivity increase in an overall process? - Throughput

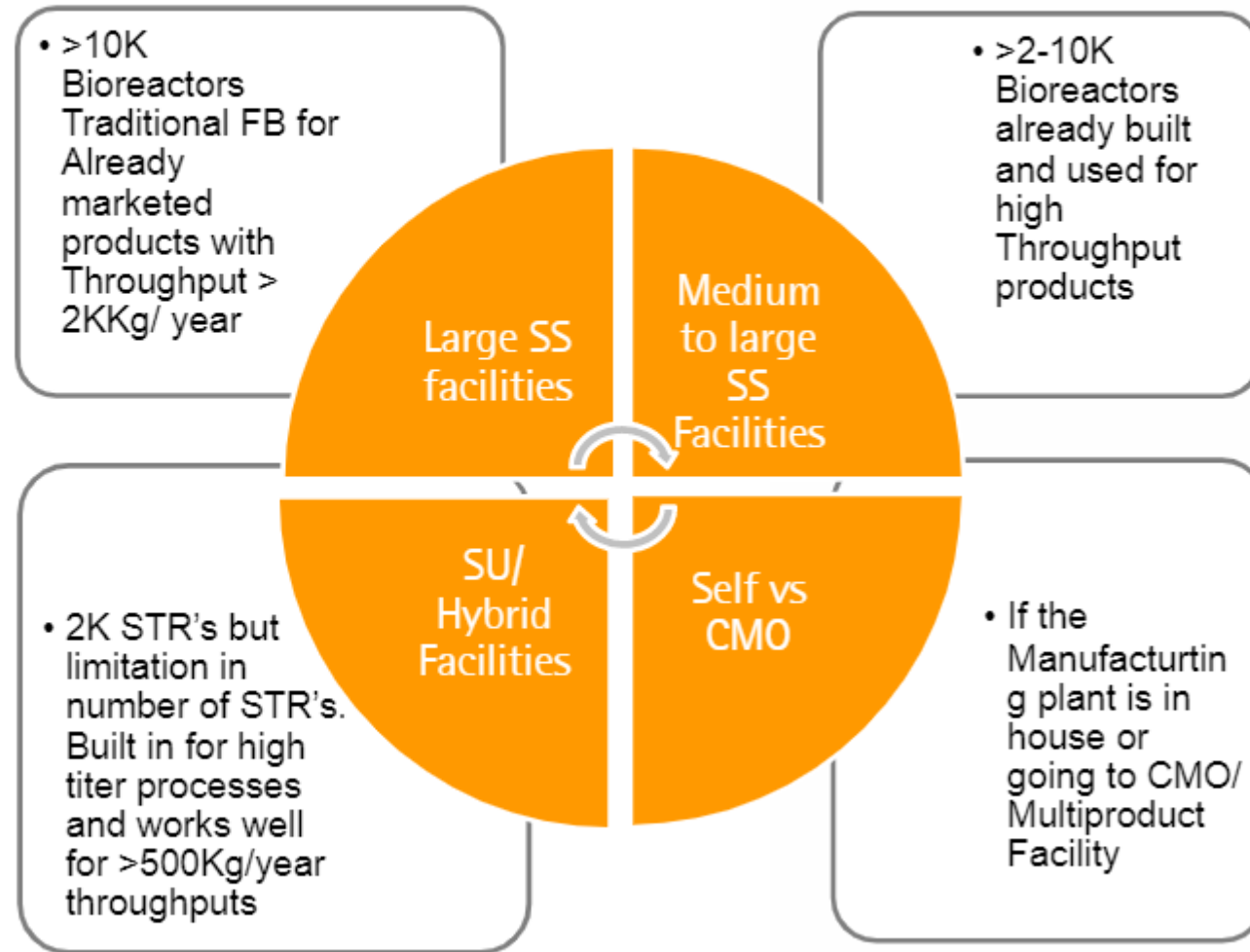
Value of increasing/improving the utilization of plant capacity? - Flexibility

Value of decoupling scale from product mass? – Scale out than scale up to produce more

Value of connecting multiple unit operations with skids? - Intensification



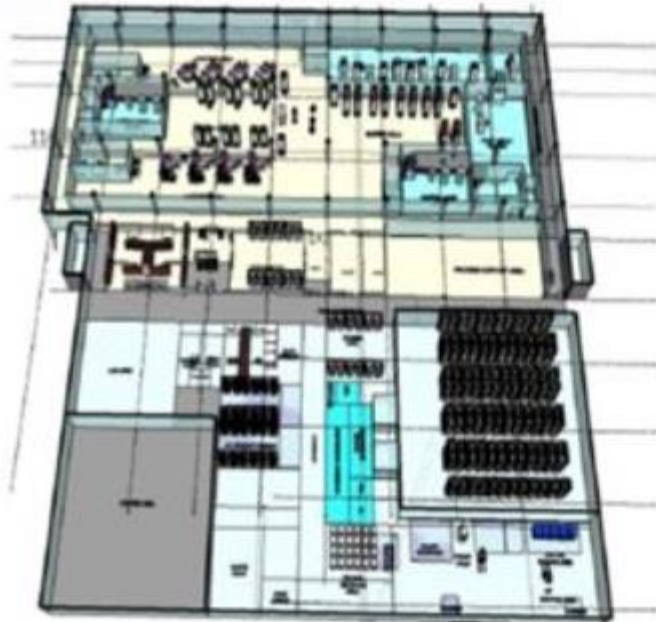
Understanding your requirement and process



High demand products	>1000Kg/year larger patient/ 24/7/ optimized
Moderate demand Products	upto 500 Kg/year moderate patient/ optimized/ flexible
Earlier phases	10-500 Kg/year short campaigns/ low Opex/ flexible

Single-Use Meets Biotech Industry Needs

The industry has understood the benefits of SU but challenges for adoption remain Specially for High-Throughput Products



Single Use Facility fit
for 6x 2,000L SUBs & Batch DSP

- Companies still investing in SS facility
- When does scaling out become obsolete and one has to scale up
- How to reduce manufacturing plant foot print
- How to build plants faster
- Can multiproduct facilities produce multiple products in an year?

Facility Goals:

- ✓ 80% less Footprint
- ✓ 75% less capital
- ✓ 50% faster plant buildup
- ✓ 60% reduction in OPEX cost

➤ Can process intensification solve the above needs and deliver??

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Fed Batch, Perfusion and Integrated Continuous Process Flow

Enzene's Integrated Continuous Processing Setup

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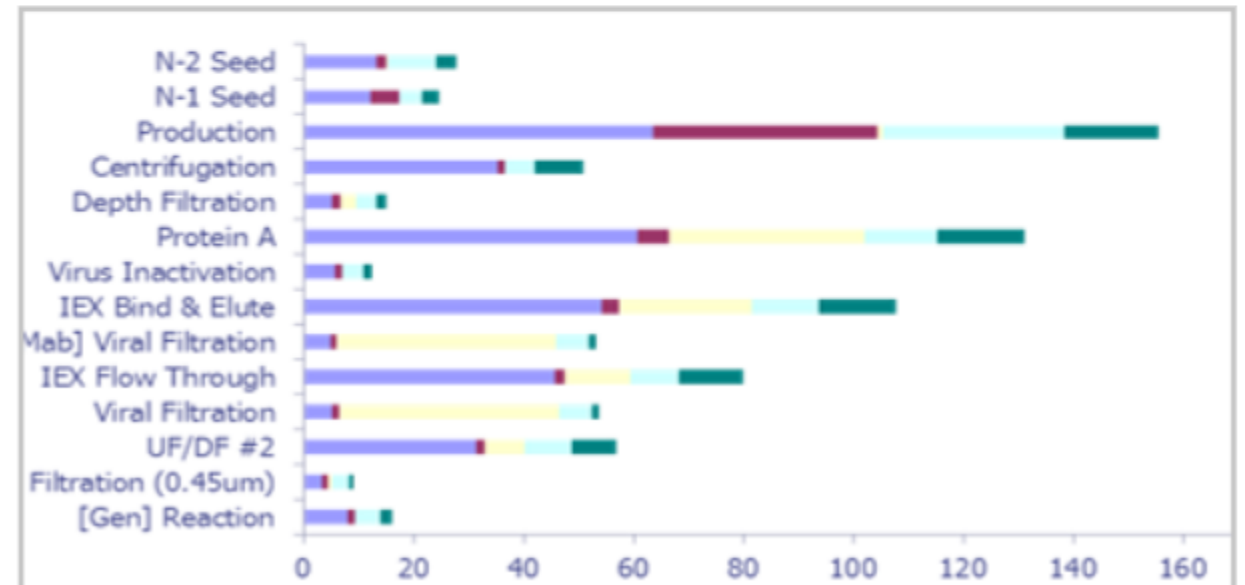
The Need for Process Modeling

Provides a holistic picture and connects all the elements to streamline decision-making

- Get the big picture – what are the variables?
- Make predictions – what will happen if...
- Evaluate alternatives
 - Which one is best?
- Cost Implications
 - Impact on cost with changes
 - Make better business decisions
 - Design or adapt to the changes in a cost effective way



Cost breakdown by unit operation
\$\$\$ / batch for a mAb process



Factors to consider while designing a process

- Facility Fit – Existing or new one
- Innovator vs Biosimilar – Investment/ Productivity
- SU or SS – Scale up or Scale out
- Throughput/year – Kg/year
- USP – DSP combinations

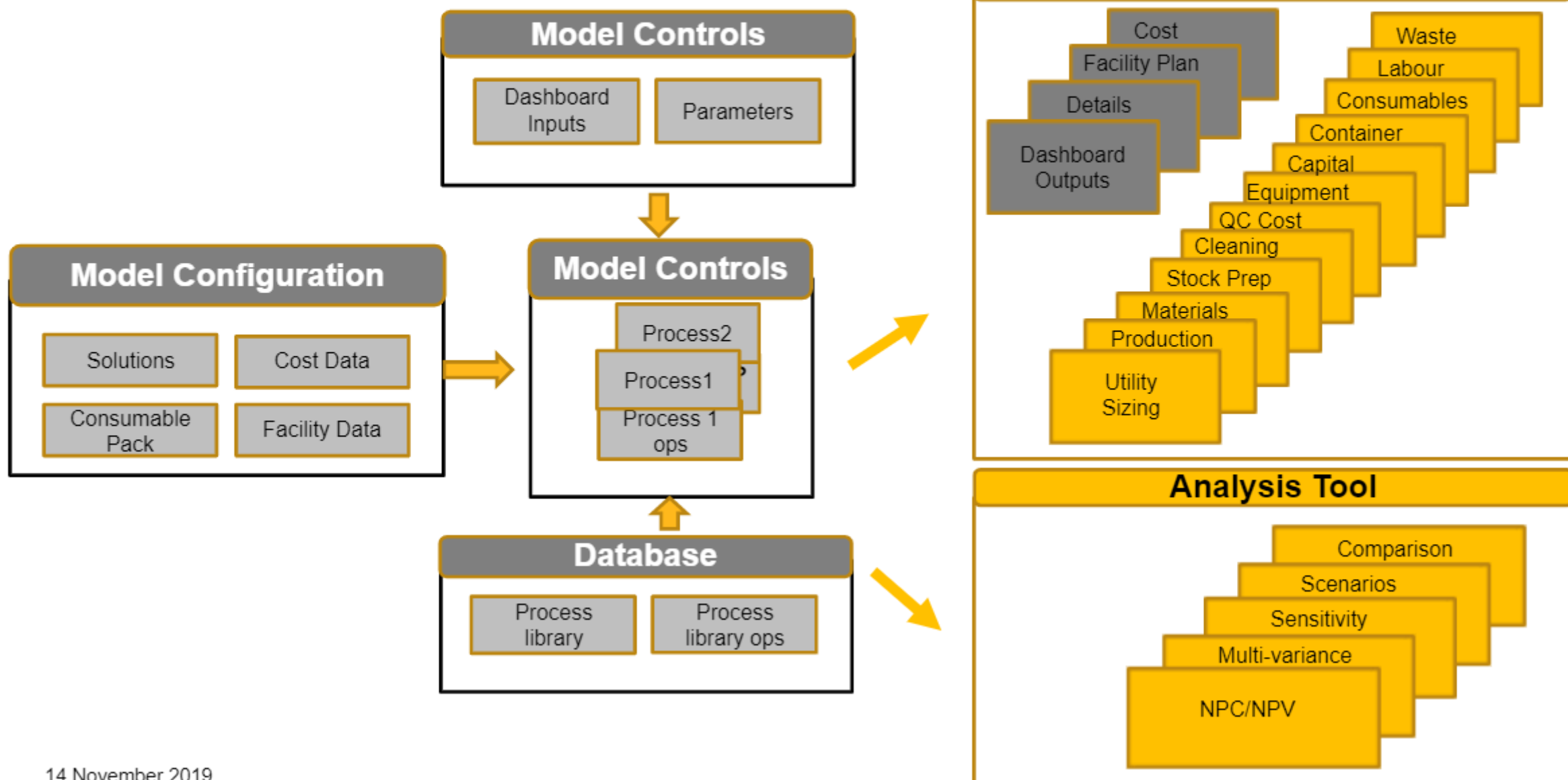
- **Various USP Combinations from Seed to Production Scale**

1. Fed Batch Strategy
2. N-1 perfusion Strategy
3. Conc. fed batch/perfusion strategy
4.

- **Various DSP Scenarios**

1. Batch
2. Multi Column
3. Cont. Harvest (w/ ATF)
4. Cont. Chrom
5. Fully Cont.
6.

Overview Biosolve



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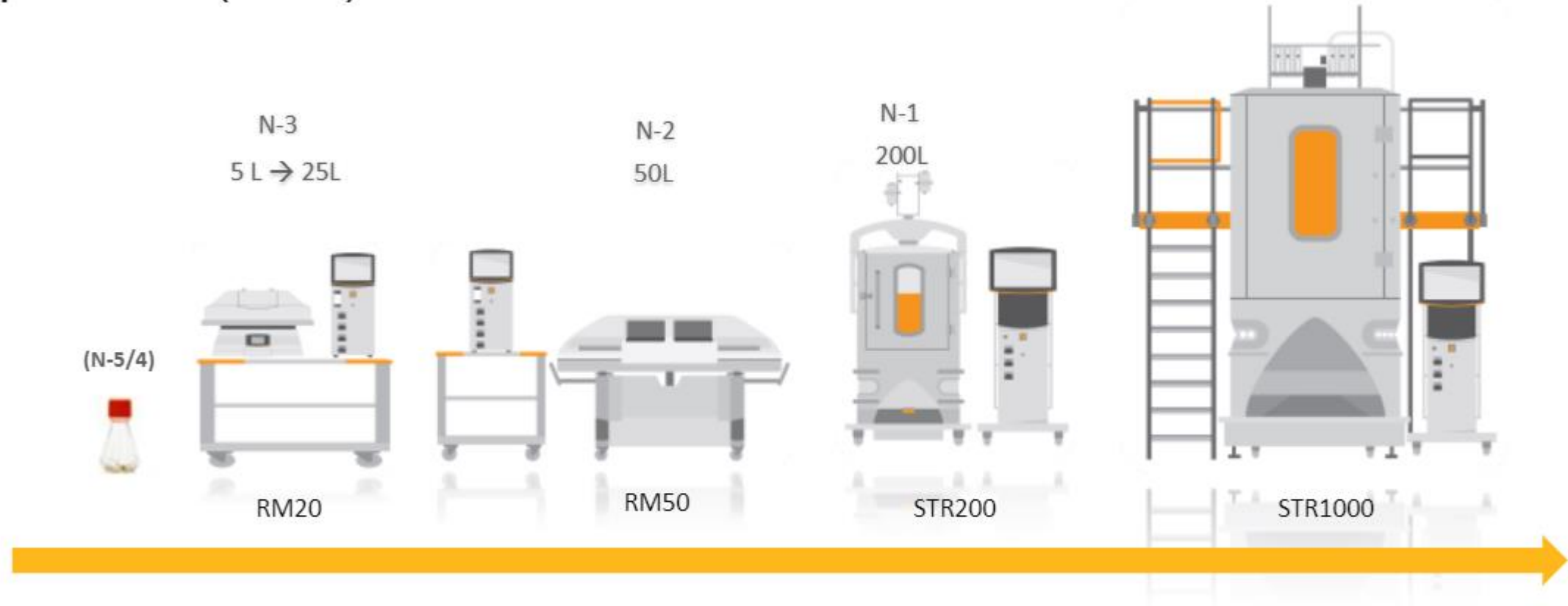
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Upstream (USP) Seed to Production: Fed Batch



Total Production STR days – 12 days

Titer – 3g/L

Total Throughput per batch = 3Kg

Upstream (USP) Seed to Production: Current Enzene Platform

(N-5/4)

0.1 L → 1L



N-2

1 L → 5L



N-1

5 L → 25L



RM20 / RM50



STR200

REPLIGEN
+ ATF

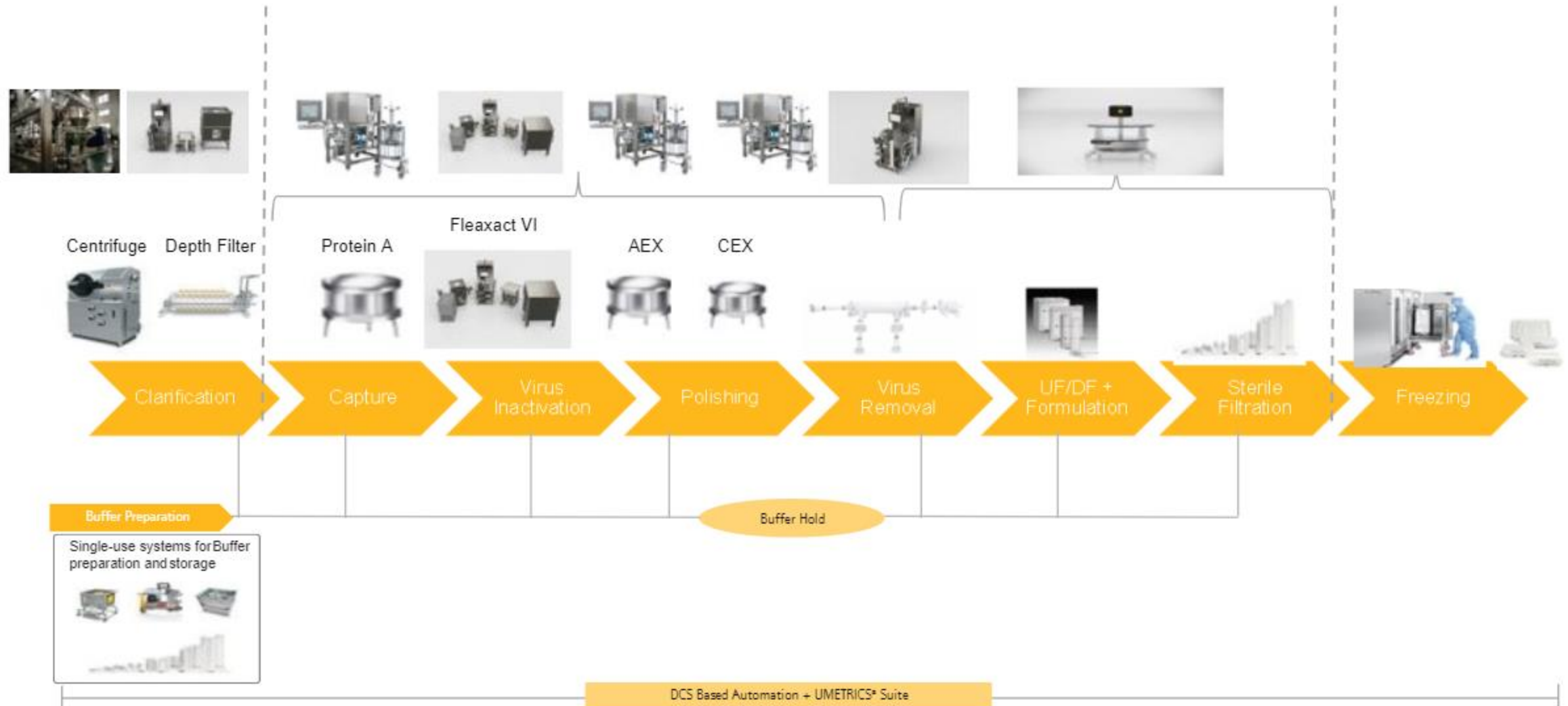
Total Production STR days – 25 days

Titer – 1.5g/L

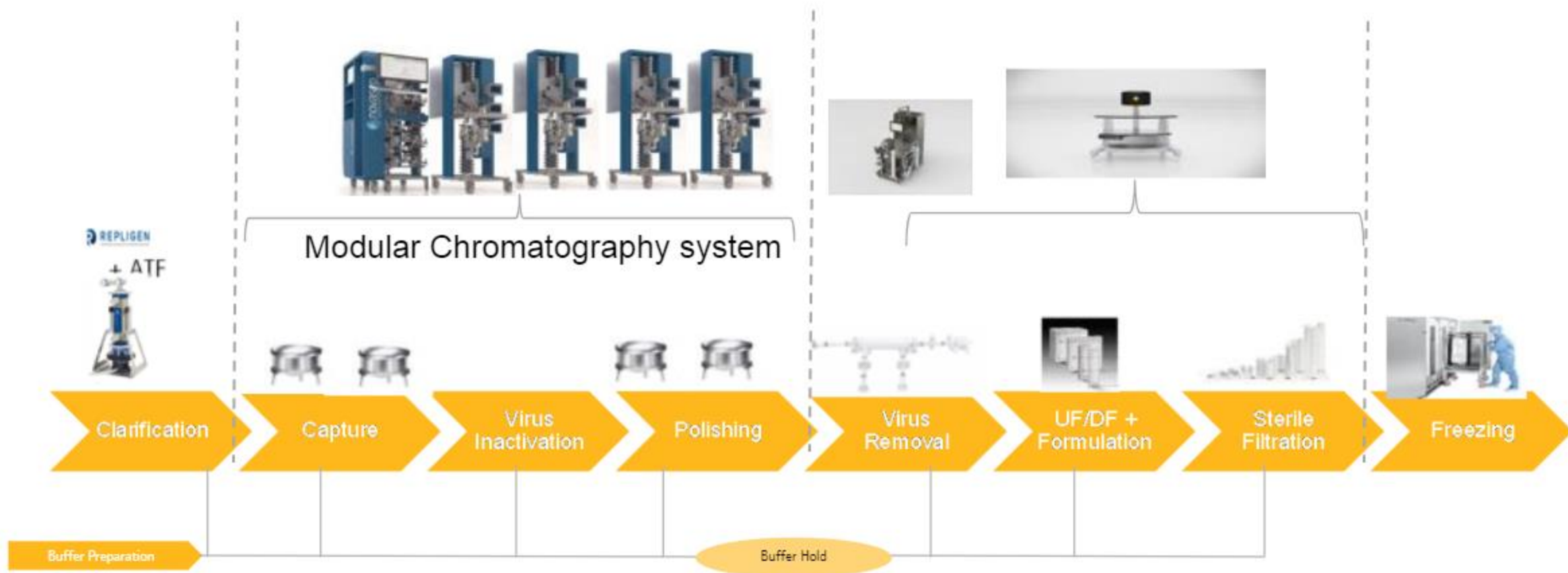
Total Throughput per batch = 6Kg

~ 5 fold reduction in footprint with double the productivity per batch

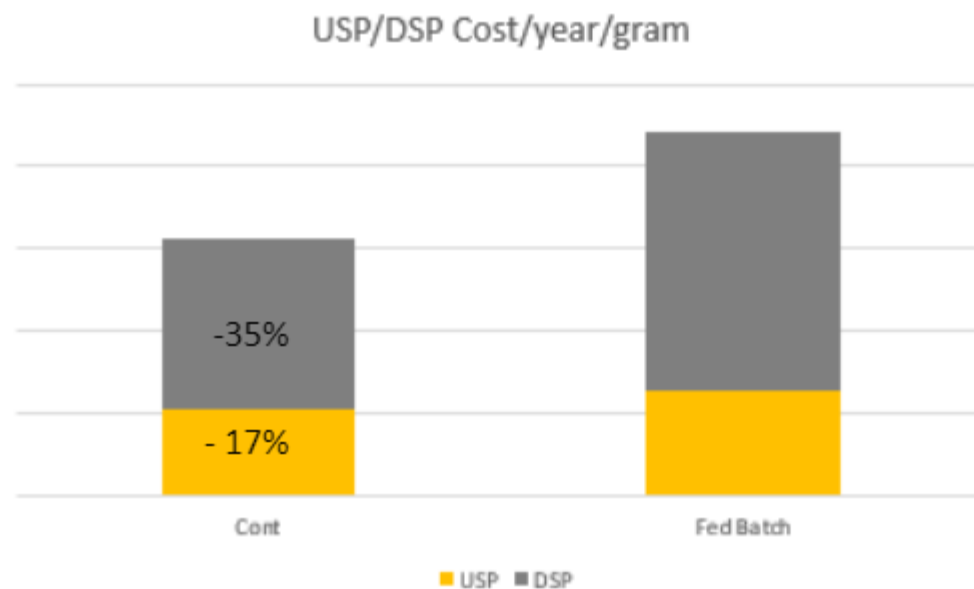
Downstream (DSP) Clarification to final Filtration – Batch DSP



Downstream (DSP) Clarification to final Filtration – Enzene Platform



Cost Analysis



~ 30% reduction in cost/ gram →

Lower cost of drug

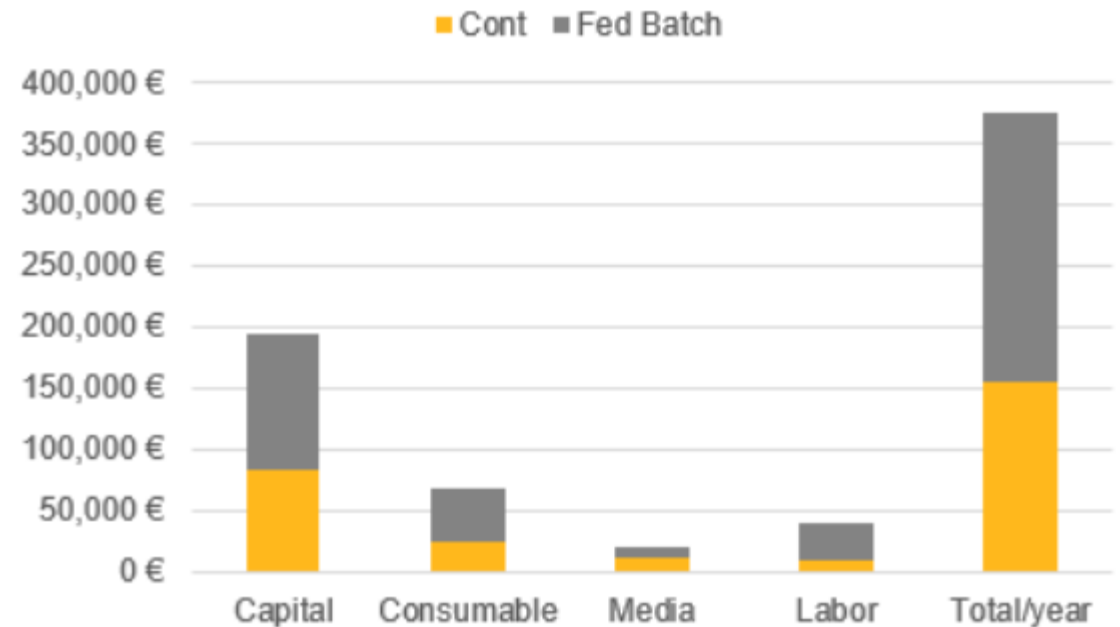
Lower USP and DSP footprint and running cost →

Modular facility with low investment and recurring cost

Cost Analysis

- Lower Overall Capital – Low investment cost for a new company/facility build
- Lower Footprint – Smaller facility – modular facility
- Lower Consumable – Running cost reduced to product low cost of drug
- Less Labor cost – Reduced recurring cost – Low cost of drug
- Media cost high – Media optimization necessary to reduce the cost of media as Perfusion process requires high media volumes

Capex/Opex distribution per year/gram



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Understanding Process and Cost investments

- Titer – Plays an important role for increase productivity
 - Media – Perfusion process requires high media volume hence media optimization and price an important factor
 - Automation
-
- How to reduce cost/g further to come in the range of \$20-30/g?

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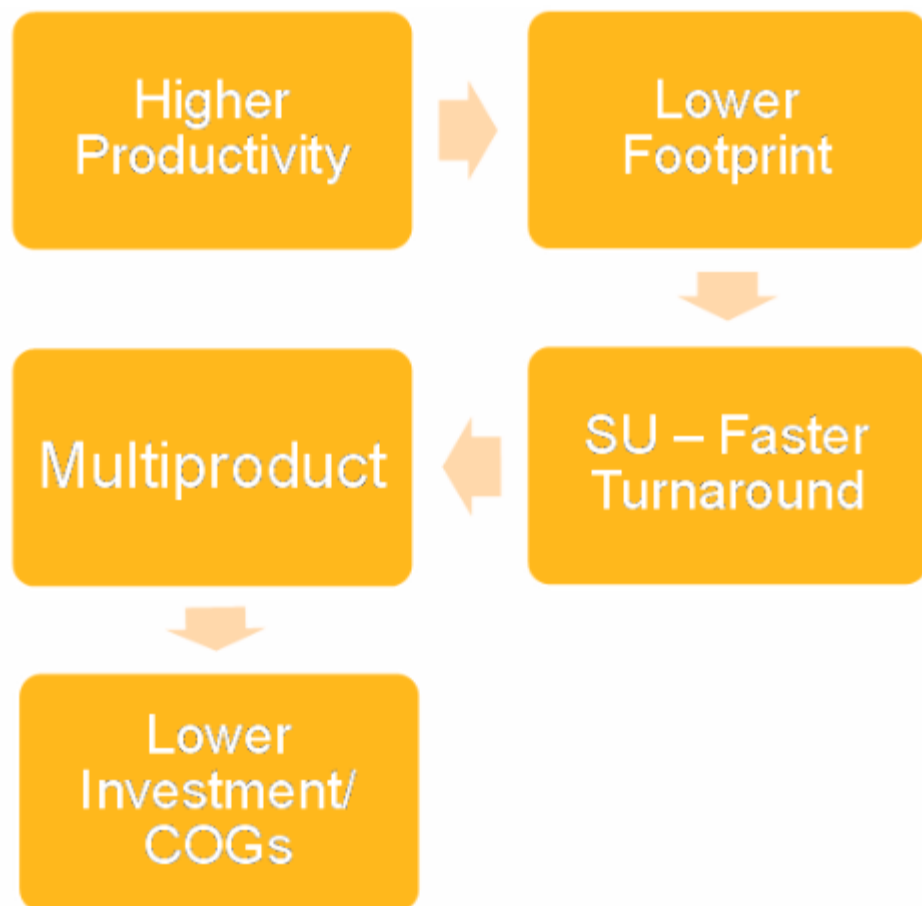
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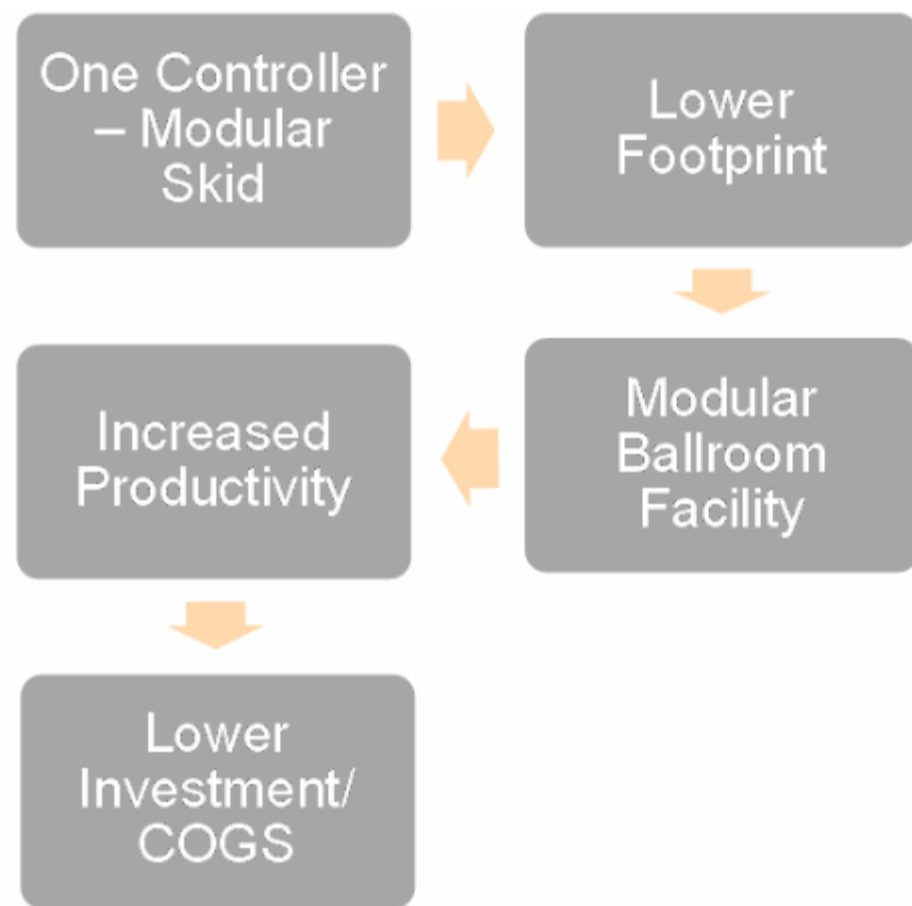


Summary

Continuous Upstream



Continuous Downstream



Contact



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Thank you!

